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# **ALGEBRA-SUMMER**

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# Contents

|          |                                            |          |
|----------|--------------------------------------------|----------|
| 0.1      | Intro                                      | 3        |
| <b>1</b> | <b>Groups</b>                              | <b>5</b> |
| 1.1      | Intro                                      | 5        |
| 1.1.1    | Exercises                                  | 5        |
| 1.2      | Dihedral Groups                            | 6        |
| 1.2.1    | Symmetry Group for an Equilateral Triangle | 6        |
| 1.2.2    | Symmetry Group for a Square                | 8        |
| 1.2.3    | Symmetry Group of regular n-gon            | 9        |
| 1.2.4    | Exercises                                  | 10       |
| 1.3      | Symmetric Groups                           | 11       |
| 1.3.1    | Cycle Notation                             | 12       |
| 1.3.2    | Exercises                                  | 15       |
| 1.4      | General Results on Groups                  | 17       |
| 1.4.1    | Subgroup Lattice                           | 18       |
| 1.4.2    | Exercises                                  | 21       |
| 1.5      | Cyclic Groups                              | 22       |
| 1.5.1    | Exercises                                  | 26       |
| 1.6      | Homomorphism and Isomorphism               | 27       |
| 1.6.1    | Exercises                                  | 31       |
| 1.7      | Cosets                                     | 32       |
| 1.7.1    | Exercises                                  | 34       |
| 1.8      | Lagrange's Theorem                         | 35       |
| 1.8.1    | Exercises                                  | 37       |
| 1.9      | Normal Subgroups and Quotient Groups       | 38       |
| 1.9.1    | Exercises                                  | 39       |

## 0.1 Intro

**Theorem 0.1.1** (Division Algoritm). Suppose  $a, b \in \mathbb{Z}$  s.t.  $b > 0$ . Then there are unique  $q, r \in \mathbb{Z}$  s.t.  $a = bq + r$  and  $0 \leq r < b$

*Proof.* Let  $a, b \in \mathbb{Z}$  s.t.  $b > 0$ . Consider  $A = \{a - bx \mid x \in \mathbb{Z}\} \cap \mathbb{N}$

By the Well Ordering Principle since  $A$  has all non-negative terms it has a minimum lets call this  $r$  and call the associated  $x \in \mathbb{Z}$ ,  $q$ .

Thus,  $r = a - bq \implies a = bq + r$ .

For contradiction, assume  $r \geq b$ .

$\implies r = b + m$  s.t.  $m \geq 0$

$\implies b + m = a - bq \implies m = a - b(q + 1) \in A \implies m < r = \text{minimum} \implies \Leftarrow$  Thus  $0 \leq r < b$

Fix  $q_1, q_2, r_1, r_2 \in \mathbb{Z}$  s.t.  $a = bq_1 + r_1 = bq_2 + r_2$  and  $0 \leq r_1, r_2 < b$ . WLOG let  $r_1 \geq r_2$

$\implies bq_1 + r_1 = bq_2 + r_2 \implies r_1 - r_2 = b(q_2 - q_1) \implies r_1 - r_2$  is a multiple of  $b$

Since  $0 \leq r_1, r_2 < b \implies 0 \leq r_1 - r_2 < b$

Since  $r_1 - r_2$  is strictly less than  $b$  the only multiple of  $b$  it could be is 0 thus  $r_1 = r_2$

$\implies q_1 = q_2$  □

**Lemma 0.1.2.** (Bezout's Lemma) Suppose  $a, b \in \mathbb{Z} \setminus \{0\}$ . Then there are  $x, y \in \mathbb{Z}$  s.t.  $ax + by = \gcd(a, b)$ .

*Proof.* Let  $a, b \in \mathbb{Z} \setminus \{0\}$ . Consider  $A = \{ax + by \mid x, y \in \mathbb{Z}\} \cap \mathbb{N}$

By the Well Ordering Principle since  $A$  has all non-negative terms it has a minimum lets call this  $d$  and call the associated  $x, y \in \mathbb{Z}$ ,  $u$  and  $v$ .

By Theorem 0.1.1  $\exists q, r \in \mathbb{Z}$  s.t.  $a = dq + r$  and  $0 \leq r < d$

Case 1:  $r = 0 \implies d \mid a$

Case 2:  $0 < r < d \implies r = a - dq \implies r = a - (au + bv)q = a(1 - uq) + b(-vq) \in A$

$\implies r < d = \text{minimum} \implies \Leftarrow$

Thus  $d \mid a$  and similarly  $d \mid b$

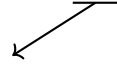
Let  $c \in \mathbb{N}$  s.t.  $c \mid a$  and  $c \mid b$

$\implies a = cm$  and  $b = cn$  for  $m, n \in \mathbb{Z}$

$d = au + bv = c(mu) + c(mv) = c(mu + mv) \implies c \mid d \implies d = \gcd(a, b)$  □

**Algorithm 0.1.3.** (Euclidean Algorithm)Find  $\gcd(272, 571)$  and  $x, y \in \mathbb{Z}$  s.t.  $272x + 571y = \gcd(272, 571)$ 

$$571 = 272 \cdot 2 + 27 \quad (1)$$



$$272 = 27 \cdot 10 + 2 \quad (2)$$



$$27 = 2 \cdot 13 + 1 \quad (3)$$

Last term before remainder is zero is gcd

$$\gcd(272, 571) = 1$$

$$2 = 272 - 27 \cdot 10$$

Use equation (2)

$$= 272(21) - 571(10)$$

$$1 = (571 - 272 \cdot 2) - 13(272(21) - 571)$$

Use equation (3) with (2)

$$= 571(131) + 272(-275)$$

# 1 Groups

## 1.1 Intro

**Definition 1.1.1.** A group is a set  $G$  together with a binary operation  $*$  :  $G \times G \rightarrow G$  s.t.

- **(Closure)** For all  $a, b \in G$  we have  $a * b \in G$
- **(Identity)** There is  $e \in G$  s.t. for all  $a \in G$   $a * e = e * a = a$
- **(Inverses)** For all  $a \in G$  there is an  $a^{-1} \in G$  s.t.  $a * a^{-1} = a^{-1} * a = e$
- **(Associativity)** For all  $a, b, c \in G$  we have  $a * (b * c) = (a * b) * c$

**Definition 1.1.2.** Given a group  $G$  and  $g \in G$  we say the order of  $g$  is  $n \in \mathbb{N}$  if  $n$  is the minimum value s.t.  $g^n = g * g * \dots * g = e$   
Write as:  $\text{ord}(g) = |g| = n$

### 1.1.1 Exercises

**Example 1.1.3.** Let  $G = \mathbb{R} \setminus \{-1\}$  and define  $a * b = a + b + ab$ . Show  $(G, *)$  is an Abelian group

- (Abelian) For all  $a, b \in G$   $a * b = a + b + ab = b + a + ba = b * a$
- (Closure)  $\checkmark$
- (Identity) For all  $a \in G$   $a * 0 = 0 * a = 0$
- (Inverses) For all  $a \in G$  let  $b = \frac{-a}{a+1}$ .

$$\text{Thus, } a * b = b * a = a + b + ab = a + b(a+1) = \frac{-a}{a+1}(a+1) = a - a = 0$$

- (Associativity) For all  $a, b, c \in G$

$$a * (b * c) = a * (b + c + bc) = a + b + c + bc + ab + ac + abc = (a + b + ab) * c = (a * b) * c$$

Therefore,  $G$  is an Abelian group

**Example 1.1.4.** Find all  $x \in \mathbb{Z}$  s.t.  $5x \equiv 4 \pmod{9}$

$$5x \equiv 4 \pmod{9}$$

$$x \equiv 8 \pmod{9}$$

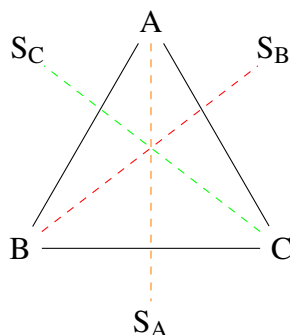
$$x = 9n + 8$$

$$2 * 5 = 10 \equiv 1 \pmod{9}$$

$$n \in \mathbb{Z}$$

## 1.2 Dihedral Groups

### 1.2.1 Rigid Motions of an Equilateral Triangle



#### Rotations

$r = 120^\circ$  rotation

$r^2 = 240^\circ$  rotation

$r^3 = 360^\circ$  rotation  $= e = 0^\circ$  rotation

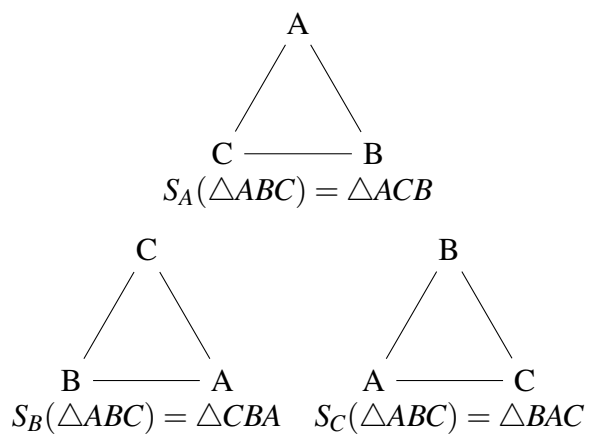
#### Reflections

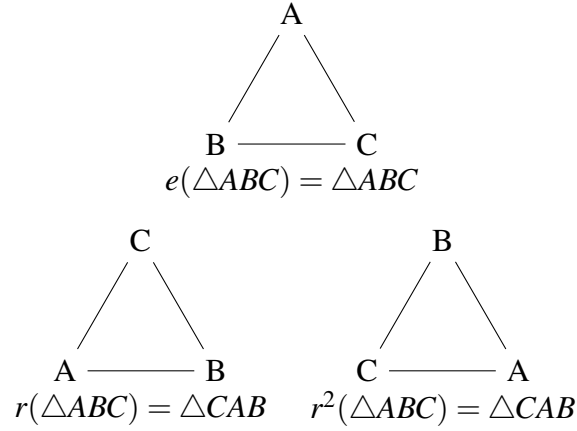
$S_A : \triangle ABC \rightarrow \triangle ACB$

$S_B : \triangle ABC \rightarrow \triangle CBA$

$S_C : \triangle ABC \rightarrow \triangle BAC$

#### Transformation Diagrams





### Operations

$$\begin{aligned}
 r \circ S_A(\triangle ABC) &= r(\triangle ACB) = \triangle BAC = S_C(\triangle ABC) & r \circ S_A &= S_C \\
 r^2 \circ S_A(\triangle ABC) &= r^2(\triangle ACB) = \triangle CBA = S_B(\triangle ABC) & r^2 \circ S_A &= S_B
 \end{aligned}$$

Now we only need  $S_A$  so lets call this  $S$

Thus we only need  $S, r, r^2, S, r, r^2, e$

Also  $S^2 = r^3 = e$

### How $S, r, r^2$ Commute

$$\begin{aligned}
 r \circ S(\triangle ABC) &= r(\triangle ACB) = \triangle BAC = S \circ r^2 \\
 r^2 \circ S(\triangle ABC) &= r^2(\triangle ACB) = \triangle CBA = S \circ r
 \end{aligned}$$

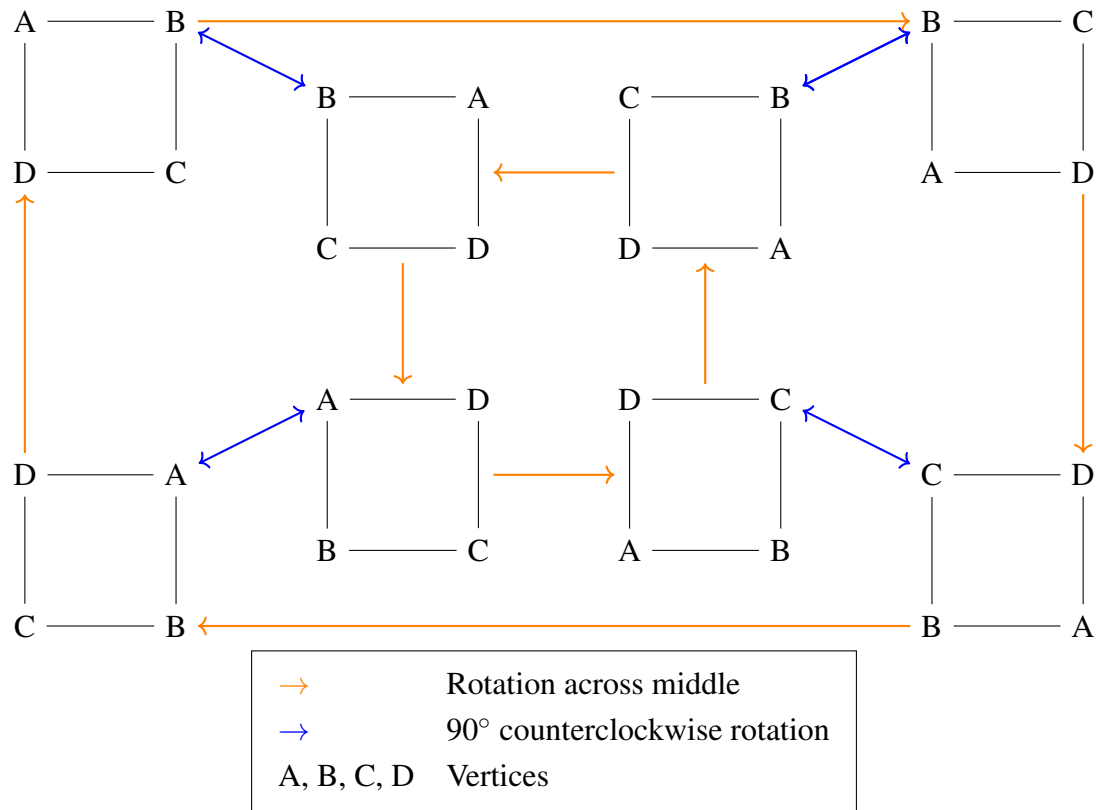
**Definition 1.2.1.** The group of symmetries of an equilateral triangle is  $D_3 = \{e, r, r^2, S, Sr, Sr^2\}$ , where  $r^3 = S^2 = e$ ,  $rS = Sr^2$

Dihedral group of order  $2 \cdot 3 = 6$

**Compact Notation**  $D_3 = \langle r, s^1 \mid r^3 = S^2 = e, rS = Sr^2 \rangle$

<sup>1</sup>The generators of  $D_3$  meaning you can create all other operations from them

## 1.2.2 Symmetry Group for a Square



### Rotations

$$\begin{aligned}
 r &= 90^\circ \text{ rotation} \\
 r^2 &= 180^\circ \text{ rotation} \\
 r^3 &= 270^\circ \text{ rotation} \\
 r^4 &= 360^\circ \text{ rotation} = e = 0^\circ \text{ rotation}
 \end{aligned}$$

### Reflections

$$s : \square ABCD \rightarrow \square BCDA$$

### How does $s, r$ Commute

From the diagram we see that  $sr = r^3s \implies rs = sr^3$

**Definition 1.2.2.** The group of symmetries of a square is  $D_4 = \{e, r, r^2, r^3, s, sr, sr^2, sr^3\}$

Dihedral group of order  $2 \cdot 4 = 8$

$$D_4 = \langle s, r \mid r^4 = s^2 = e, sr = r^3s \rangle$$

### 1.2.3 Symmetry Group of regular n-gon

Rotation:  $r = \frac{360^\circ}{n} \circlearrowleft (e, r, r^2, \dots, r^{n-1})$

Reflection:  $s = \text{reflection across middle } (s, sr, sr^2, \dots, sr^{n-1})$

**Claim:**  $rs = sr^{n-1}$ .

*Proof.* We only need 3 points since our shape is defined by the order of those 3 points

$$r \circ s(n, 1, 2) = r(2, 1, n) = (1, n, n-1)$$

$$s \circ r^{n-1}(n, 1, 2) = s(n-1, n, 1) = (1, n, n-1)$$

$$\text{Thus } rs = sr^{n-1}$$

□

**Definition 1.2.3.** The dihedral group  $D_n$  is the symmetry group of a regular n-gon.

$$D_n = \langle s, r \mid r^n = s^2 = e, rs = sr^{n-1} \rangle$$

**Claim:** For all  $1 \leq m \leq n-1$  we have  $r^m s = sr^{n-m}$ .

*Proof.* Base Case:  $m = 1 \implies rs = sr^{n-1} \checkmark$

Inductive Step: Assume for  $k \geq 1$  the statement is true

Consider:

$$r^{k+1}s = rr^k s$$

$$= rsr^{n-k}$$

$$= sr^{n-1}r^{n-k}$$

$$= sr^n r^{n-(k+1)}$$

$$= sr^{n-(k+1)}$$

Apply inductive hypothesis on  $r^k s$

□

### 1.2.4 Exercises

**Example 1.2.4.** Simplfy

- $(sr^5) r (sr^3)$  in  $D_4$

$$= (sr)r(rs)$$

$$= e$$

- $(sr^4) (sr^2)$  in  $D_6$

$$= (r^2s)(sr^2) \quad \text{Use } r^m s = sr^{n-m}$$

$$= r^4$$

- $r^3 (sr)$  in  $D_4$

$$= r^3(r^3s)$$

$$= r^2s$$

**Example 1.2.5.** The Cayley table of the  $D_3$  can be written

|        | $e$    | $r$    | $r^2$  | $s$    | $rs$   | $r^2s$ |
|--------|--------|--------|--------|--------|--------|--------|
| $e$    | $e$    | $r$    | $r^2$  | $s$    | $rs$   | $r^2s$ |
| $r$    | $r$    | $r^2$  | $e$    | $rs$   | $r^2s$ | $s$    |
| $r^2$  | $r^2$  | $e$    | $r$    | $r^2s$ | $s$    | $rs$   |
| $s$    | $s$    | $r^2s$ | $rs$   | $e$    | $r^2$  | $r$    |
| $rs$   | $rs$   | $s$    | $r^2s$ | $r$    | $e$    | $r^2$  |
| $r^2s$ | $r^2s$ | $rs$   | $s$    | $r^2$  | $r$    | $e$    |

### 1.3 Symmetric Groups

**Definition 1.3.1.** Given a set  $X$ , a permutation of  $X$  is a bijection  $X \rightarrow X$   
 $S_X = \{ \sigma : X \rightarrow X \mid \sigma \text{ is a bijection} \}$

**Lemma 1.3.2.** For all sets  $X \neq \emptyset$ ,  $S_X$  is a group

*Proof.*

- Closure: Suppose  $\sigma, \mu \in S_X$   
 Consider  $\sigma \circ \mu$  since it is a composition of bijections it is also a bijection.  
 Also functions go from  $X \rightarrow X$  thus  $\sigma \circ \mu$  is the same.  
 Therefore, the composition is a bijection that is mapped from  $X \rightarrow X$  thus  $\sigma \circ \mu \in S_X$
- Identity: Consider  $e : X \rightarrow X$  s.t.  $e(x) \mapsto x$   
 Thus  $e \in S_X$  and for all  $\sigma \in S_X$   $\sigma \circ e(x) = \sigma(x)$   
 Thus  $\sigma \circ e = \sigma$ . Also, it follows that  $e \circ \sigma = \sigma$
- Inverses: Since  $S_X$  has only bijective functions all bijective functions have inverses.
- Associativity: We have associativity by the fact that function composition has associativity.

□

**Definition 1.3.3.** The permutation group on  $n$ -objects  $S_n$  is the  $S_X$  with  $X = [n]$   
 Observe  $|S_n| = n!$

**Example 1.3.4.**  $\sigma, \mu \in S_4$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \mu = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix}$$

$$\sigma^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \mu^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix}$$

$$\sigma\mu = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \mu\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix}$$

### 1.3.1 Cycle Notation

**Example 1.3.5.**  $\sigma, \mu \in S_5$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 2 & 5 & 1 \end{pmatrix} \mu = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 2 & 1 & 3 \end{pmatrix}$$

$$\sigma = (13245) \mu = (14)(253)$$

Example Calculations:

$$\sigma\mu = (13245)(14)(253)$$

$$= (152)(34)$$

$$\mu\sigma = (14)(253)(13245)$$

$$= (12)(354)$$

$$\mu^2 = (14)(253)(14)(253)$$

$$= (14)^2(253)^2 = (235)$$

**Lemma 1.3.6.** *Disjoint cycles are commutative*

*Proof.* Let  $\alpha, \beta$  be disjoint cycles of  $[n]$

Let  $\alpha = (a_1, a_2, \dots, a_k)$ , and  $\beta = (b_1, b_2, \dots, b_j)$

Let  $x \in [n]$

Consider:  $\alpha\beta(x), \beta\alpha(x)$

If  $x \in \alpha$  then  $x \notin \beta \implies \beta(x) = x \implies \alpha\beta(x) = \alpha(x) = a_i$   
 $\implies \beta\alpha(x) = \beta(a_i) = a_i$  since  $a_i \notin \beta$

Use similar reasoning for if  $x \in \beta$

$\implies \alpha\beta = \beta\alpha$

□

**Lemma 1.3.7.** *Every  $\sigma \in S_n$  can be written as disjoint cycles*

**Lemma 1.3.8.** *The order of an  $m$ -cycle is  $m$*

**Lemma 1.3.9.** Suppose  $G$  is a group and  $\text{ord}(g) = n$ . Then  $g^m = e \iff n \mid m$

*Proof.* Let  $G$  be a group and  $\text{ord}(g) = n$

( $\Leftarrow$ )

Assume  $n \mid m$

$$\implies m = nk, k \in \mathbb{Z}$$

$$g^m = g^n k = (g^n)^k = (e)^k = e$$

( $\Rightarrow$ )

Assume  $g^m = e$  and use the division algorithm (0.1.1) to write  $m = nq + r$  s.t.  $0 \leq r < n$

$$e = g^m = g^{nq+r} = (g^n)^q g^r = g^r \implies r = 0 \implies n \mid m$$

□

**Theorem 1.3.10.** If  $\sigma_1, \sigma_2, \dots, \sigma_k$  are disjoint cycles of length  $a_1, a_2, \dots, a_k$  and  $\sigma = \sigma_1 \sigma_2 \dots \sigma_k$ . Then  $\text{ord}(\sigma) = \text{lcm}(a_1, a_2, \dots, a_k)$ .

*Proof.* Let  $m = \text{ord}(\sigma)$ ,  $l = \text{lcm}(a_1, a_2, \dots, a_k)$

Since  $l$  is a lcm it is also a multiple so  $l = a_j q_j$  for  $1 \leq j \leq k$  where  $q_j \in \mathbb{Z}$

$$\sigma^l = \sigma_1^l \sigma_2^l \dots \sigma_k^l$$

We can do this because of Lemma 1.3.6

$$= (\sigma_1^{a_1})^{q_1} \dots (\sigma_k^{a_k})^{q_k}$$

$$= (1)$$

Apply Lemma 1.3.8

$$\implies m \mid l$$

Also,  $(1) = \sigma^m = \sigma_1^m \dots \sigma_k^m$  since they are disjoint none of them are inverses of each other

$$\implies \sigma_1^m = \dots = \sigma_k^m = (1)$$

$$(\text{Lemma 2.1.9}) \implies a_1 \mid m \dots a_k \mid m \implies l \mid m$$

Thus, since  $l \mid m$  and  $m \mid l$   $m = l$

□

**Example 1.3.11.**

- Find an element of order 12 in  $S_7$   
 $(1234)(567) \in S_7$  by Theorem 1.3.10  $\implies$  the order is  $3 \cdot 4 = 12 = \text{lcm}(3,4)$
- Possible orders of elements in  $S_5$ 
  - 1:  $(1)$ ,  $5=1+1+1+1+1$
  - 2:  $(12)$ ,  $5=2+1+1+1+1$  or  $(12)(34)$ ,  $5=2+2+1$
  - 3:  $(123)$ ,  $5=3+1+1$
  - 4:  $(1234)$ ,  $5=4+1$
  - 5:  $(12345)$ ,  $5=5$
  - 6:  $(12)(345)$ ,  $5=2+3$

### 1.3.2 Exercises

**Example 1.3.12.** Write the following as the product of disjoint cycles and find the order.

- $(1358)(378)(125) \in S_8$   
 $(128537)$  order=6
- $(1264)^{-1}(364)^2 \in S_6$   
 $= (1462)(346)$   
 $= (142)(36)$  order=6

**Example 1.3.13.** Find all Possible orders of elements in  $S_7$

- 1:  $(1)$ ,  $7=1+1+1+1+1+1+1$
- 2:  $(12)$ ,  $7=2+1+1+1+1+1$  or  $(12)(34)$ ,  $7=2+2+1+1+1$  or  $(12)(34)(56)$ ,  $7=2+2+2+1$
- 3:  $(123)$ ,  $7=3+1+1+1+1$  or  $(123)(456)$ ,  $7=3+3+1$
- 4:  $(1234)$ ,  $7=4+1+1+1$  or  $(1234)(56)$ ,  $7=4+2+1+1$
- 5:  $(12345)$ ,  $7=5+1+1$
- 6:  $(123456)$ ,  $7=6+1$  or  $(123)(45)$ ,  $7=3+2+1+1$ , or  $(123)(45)(67)$ ,  $7=3+2+2+1$
- 7:  $(1234567)$ ,  $7=7$
- 10:  $(12345)(67)$ ,  $7=5+2$
- 12:  $(1234)(567)$ ,  $7=4+3$

**Example 1.3.14.** What is the largest possible order of an element in  $S_{10}$ ?

| cycle type  | order |
|-------------|-------|
| (5,1,1,1,1) | 5     |
| (5,2,1,1,1) | 10    |
| (5,2,2,1)   | 10    |
| (5,3,1,1)   | 15    |
| (5,3,2)     | 30    |
| (5,4,1)     | 20    |
| (5,5)       | 5     |
| (6,1,1,1,1) | 6     |
| (6,2,1,1)   | 6     |
| (6,3,1)     | 6     |
| (6,4)       | 12    |
| (7,1,1,1)   | 7     |
| (7,2,1)     | 14    |
| (7,3)       | 21    |
| (8,1,1)     | 8     |
| (8,2)       | 8     |
| (9,1)       | 9     |
| (10)        | 10    |

Max=30

## 1.4 General Results on Groups

**Lemma 1.4.1.** *Suppose  $G$  is a group. Then*

- (1) *The identity is unique*
- (2) *Inverses are unique*
- (3) *For every  $a, b, c \in G$  there is a unique  $x \in G$  s.t.  $ax = b$*
- (4) *If  $ab = ac$  then  $b = c$ , similarly if  $xy = zy$  then  $x = z$*
- (5)  $(ab)^{-1} = b^{-1}a^{-1}$
- (6) *If  $(ab)^2 = a^2b^2$  then  $ab = ba$*
- (7) *If  $\text{ord}(g) = 2$  for all  $g \in G$  then  $G$  is abelian*

• *Proof.* (3) Let  $ax = b \implies x = a^{-1}b$

Suppose  $x, y \in G$  s.t.  $ax = b$  and  $ay = b$

$$ax = ay \implies x = y \quad \square$$

• *Proof.* (6) Assume  $(ab)^2 = a^2b^2$

$$ab = (a^{-1}a)(ab)(bb^{-1}) = a^{-1}a^2b^2b^{-1} = a^{-1}(ab)^2b^{-1} = (a^{-1}a)(ba)(bb^{-1}) = ba \quad \square$$

• *Proof.* Assume  $\text{ord}(g) = 2$  for all  $g \in G$

$$\implies g = g^{-1} \text{ for all } g \in G$$

Let  $a, b \in G$ . Consider  $ab$

$$ab = (ab)^{-1} = b^{-1}a^{-1} = ba \quad \square$$

**Definition 1.4.2.** Given a group  $G$  and a subset  $H \subseteq G$ , we say  $H$  is a subgroup  $H \leq G$  if

- The identity of  $G$  is in  $H$
- For all  $h \in H, h^{-1} \in H$
- For all  $a, b \in H, ab \in H$

**Theorem 1.4.3.** Suppose that  $G$  is a group and  $\emptyset \neq H \subseteq G$   
 then  $H \leq G \iff$  for all  $a, b \in H, ab^{-1} \in H$

*Proof.*

( $\implies$ ) Assume  $H \leq G$

Let  $a, b \in H$

By Definition 1.4.2  $\implies b^{-1} \in H \implies ab^{-1} \in H$

( $\impliedby$ ) Assume for all  $a, b \in H, ab^{-1} \in H$

- (Identity) Consider  $a \in H$

$$\implies aa^{-1} = e \in H$$

- (Inverses)  $\checkmark$

- (Closure) Consider  $a, b^{-1} \in H$

$$\implies ab \in H$$

□

### 1.4.1 Subgroup Lattice

#### Subgroup of Order 4

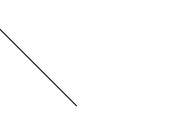
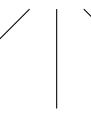
$$\mathbb{Z}_4 = \{0, 1, 2, 3\}$$



$$\begin{array}{c} | \\ 0 = \{0\} \end{array}$$

$$0 = \{0\}$$

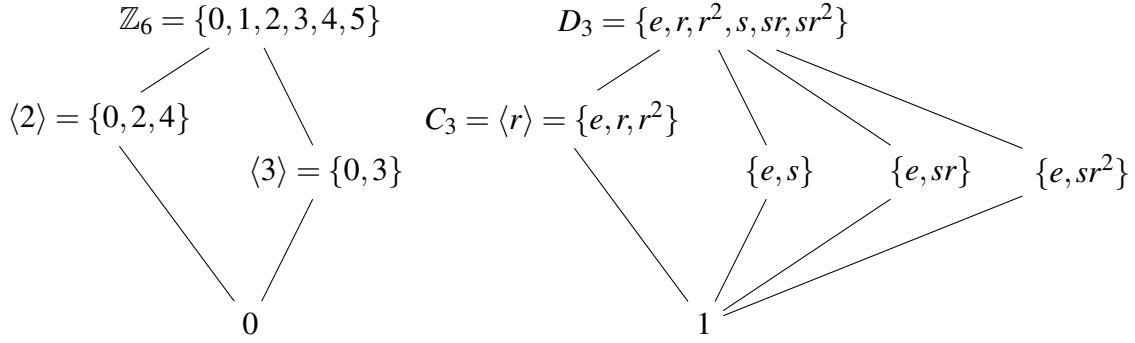
$$\mathbb{Z}_2 \times \mathbb{Z}_2 = \{(0, 0), (1, 0), (0, 1), (1, 1)\}$$



$$\begin{array}{c} \diagdown \quad | \quad \diagup \\ 0 = \{(0, 0)\} \end{array}$$

$$0 = \{(0, 0)\}$$

### Subgroup of Order 6



**Definition 1.4.4.** Given a group  $G$  and a subset  $S \subseteq G$

- The centralizer of  $S$  in  $G$ :  $C_G(S) = \{g \in G \mid gs = sg, \forall s \in S\}$
- The center of  $G$ :  $(C_G(G)) = Z(G) = \{z \in G \mid gz = zg, \forall g \in G\}$

**Lemma 1.4.5.**  $C_G(S)$  and  $Z(G)$  are subgroups of  $G$

*Proof.* Both contain  $e$  so they are nonempty

Let  $a, b \in C_G(S)$  and  $s \in S$

Consider:

$$\begin{aligned}
 as &= sa \\
 sab^{-1} &= asb^{-1} \\
 &= a(b^{-1}s)b \\
 &= (ab^{-1})s
 \end{aligned}$$

Thus  $ab^{-1} \in C_G(S)$

Thus, by Theorem 1.4.3  $C_G(S)$  is a subgroup

$\implies Z(G) \leq G$  since the center is a special case of the centralizer

□

**Example 1.4.6.** Centralizers in  $D_4$ 

- $C_{D_4}(e) = D_4 = C_{D_4}(r^2)$  <sup>2</sup>
- $C_{D_4}(r) = e, r, r^2, r^3 = C_{D_4}(r^3)$
- $C_{D_4}(s) = e, s, r^2, sr^2 = C_{D_4}(sr^2)$
- $C_{D_4}(sr) = e, sr, r^2, sr^3 = C_{D_4}(sr^3)$
- $Z(D_4) = e, r^2$

---

<sup>2</sup>Since  $r^2s = sr^2$  and  $r^2r = rr^2$

### 1.4.2 Exercises

**Example 1.4.7.** Prove that if  $G$  is abelian if and only if for all  $a, b \in G$  we have  $(ab)^{-1} = a^{-1}b^{-1}$

*Proof.* ( $\implies$ ) Assume  $G$  is abelian and let  $a, b \in G$

$$\implies (ab)^{-1} = b^{-1}a^{-1} = a^{-1}b^{-1}$$

( $\impliedby$ ) Assume for all  $a, b \in G$  we have  $(ab)^{-1} = a^{-1}b^{-1}$

Consider:  $a^{-1}$  and  $b^{-1}$

$$ba = (a^{-1}b^{-1})^{-1} = ab$$

□

**Example 1.4.8.** Suppose  $G$  is a group of order 6 containing 7 and 19 where the operation is multiplication modulo 36. List all the elements of  $G$ .

- $7 \times 7 = 49 = 13$
- $19 \times 19 = 361 = 1$
- $7 \times 19 = 133 = 25$
- $13 \times 19 = 247 = 31$

$$G = \{1, 7, 13, 19, 25, 31\}$$

**Example 1.4.9.** Suppose that  $G$  is a group such that for all  $a, b, c, d \in G$  such that if

$$axb = cxd$$

then  $ab = cd$ . Show that  $G$  is abelian

*Proof.* Consider:  $a, b \in G$

$$aa^{-1}b = ba^{-1}a \implies ab = ba$$

□

## 1.5 Cyclic Groups

**Definition 1.5.1.** Given a group  $G$  and an element  $g \in G$ , the cyclic subgroup generated by  $g$  is  $\langle g \rangle = \{g^m \mid m \in \mathbb{Z}\} = \{\dots, g^{-2}, g^{-1}, e, g, g^2, \dots\}$

We say  $G$  is a cyclic group if there is a  $g \in G$  s.t.  $G = \langle g \rangle$

$$\text{ord}(g) = |\langle g \rangle|$$

### Cyclic subgroups of $\mathbb{Z}$

#### Example 1.5.2.

- $\langle 2 \rangle = \{2m \mid m \in \mathbb{Z}\} = \{\dots, -4, -2, 0, 2, 4, \dots\} = 2\mathbb{Z}$
- $\langle n \rangle = \{nm \mid m \in \mathbb{Z}\} = \{\dots, -2n, -n, 0, n, 2n, \dots\} = n\mathbb{Z}$

### Cyclic subgroups of $\mathbb{Q}^\times$

#### Example 1.5.3.

- $\langle -1 \rangle = \{-1, 1\}$
- $\langle 2 \rangle = \{2^m \mid m \in \mathbb{Z}\}$

### Cyclic subgroups of $\mathbb{C}^\times$

#### Example 1.5.4.

$$\langle i \rangle = \{i, -1, -i, 1\}$$

### Cyclic subgroups of $\mathbb{Z}_{12}$

#### Example 1.5.5.

- $\mathbb{Z}_{12} = \langle 1 \rangle = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$
- $\langle 2 \rangle = \{0, 2, 4, 6, 8, 10\}$
- $\langle 4 \rangle = \{0, 4, 8\}$
- $\langle 3 \rangle = \{0, 3, 6, 9\}$

### Cyclic subgroups of $U_9 = \{1, 2, 4, 5, 7, 8\}$

#### Example 1.5.6.

- $\langle 1 \rangle = \{1\}$
- $\langle 8 \rangle = \{8, 64\} = \{1, 8\}$
- $\langle 4 \rangle = \{4, 16, 64\} = \{1, 4, 7\}$
- $\langle 2 \rangle = \{2, 4, 8, 16, 32, 64\} = \{1, 2, 4, 5, 7, 8\} = U_9$

### Cyclic subgroups of $D_n$

#### Example 1.5.7.

- $\langle r \rangle = \{e, r, r^2, \dots, r^{n-1}\} = Z_n^3 = C_n^4$
- $\langle s \rangle = \{e, s\}$
- $\langle sr \rangle = \{e, sr\}$

### Cyclic subgroups of $S_5$

#### Example 1.5.8.

$$\langle (12)(345) \rangle^5 = \{(1), (12)(345), (354), (12), (345), (12)(354)\}$$

**Theorem 1.5.9.** Suppose  $g \in G$  with  $\text{ord}(g) = n$  and let  $m \in \mathbb{N}$  then,  $\langle g^m \rangle = \langle g^{\gcd(m,n)} \rangle$

*Proof.* Set  $d = \gcd(m, n)$ .

Take  $x \in \langle g^m \rangle \implies x = (g^m)^k = g^{mk}$  for some  $k \in \mathbb{Z}$

Also,  $m = dq$  for some  $q \in \mathbb{Z}$  since  $d$  is a divisor of  $m$

$x = g^{dqk} = (g^d)^{qk} \in \langle g^d \rangle$  so  $\langle g \rangle \subseteq \langle g^d \rangle$

By Lemma 0.1.2 there exists  $r, s \in \mathbb{Z}$  s.t.  $mr + ns = d$

Take  $x \in \langle g^d \rangle \implies x = g^{dk}$  for some  $k \in \mathbb{Z}$

$g^{dk} = g^{k(mr+ns)} = g^{mkr}(g^n)^{ks} = g^{mkr} \in \langle g^m \rangle$  so  $\langle g^d \rangle \subseteq \langle g^m \rangle$

Thus  $\langle g^m \rangle = \langle g^d \rangle$

□

<sup>3</sup>not  $\mathbb{Z}$

<sup>4</sup>Multiplicative cyclic group of order  $n$

<sup>5</sup>By Theorem 1.3.10 the  $\text{lcm}(2,3)=6=\text{order}$  and thus this cyclic group will have 6 elements

**Theorem 1.5.10** (Fundamental Theorem of Cyclic Groups).

- I. Every subgroup of a cyclic group is cyclic
- II. If  $G$  is a cyclic group of order  $n$  and  $d \mid n$  then there is a unique subgroup  $H \leq G$  with  $|H| = d$

I. *Proof.* Suppose  $G = \langle g \rangle$  and  $H \leq G$

Case 1:  $H = \{e\} = \langle e \rangle \checkmark$

Case 2:  $H \neq \{e\}$ . Thus,  $H = \{\dots, g^{-m_2}, g^{-m_1}, e, g^{m_1}, g^{m_2}, \dots\}$  s.t.  $m_i \in \mathbb{N}$

By the Well Ordering Principle, there exists a minimum  $m_i$  call this  $m$

Since  $g^m \in H \implies (g^m)^n \in H \ \forall n \in \mathbb{Z} \implies \langle g^m \rangle \subseteq H$

Take  $h \in H$ , write  $h = g^k$  for  $k \in \mathbb{Z}$

By the division algorithm (Thm. 0.1.1), there exists a  $q, r \in \mathbb{Z}$  s.t.

$k = mq + r$  with  $0 \leq r < m$

$g^r = (g^k)(g^{-mq})$  thus since  $g^k \in H$  and  $g^{-mq} \in \langle g^m \rangle \subseteq H \implies g^r \in H$

Since  $0 \leq r < m$  and  $m$  is the minimum  $\implies r = 0$

Thus  $h = (g^m)^q \in \langle g^m \rangle \implies H \subseteq \langle g^m \rangle$

$\implies H = \langle g^m \rangle \implies H$  is cyclic □

II. *Proof.* Assume  $G$  is a cyclic group of order  $n$  and  $d \mid n$

Let  $G = \langle g \rangle$

Existence: Consider  $\langle g^{\frac{n}{d}} \rangle$  this has  $d$  elements so  $\checkmark$

Uniqueness: Let  $H \leq G$ ,  $|H| = d$  by Theorem 1.5.10.I

$\implies H = \langle g^m \rangle$  by Theorem 1.5.9  $\implies H = \langle g^{\gcd(m,n)} \rangle$

Thus  $|H| = d = \frac{n}{\gcd(m,n)} \implies \gcd(m,n) = \frac{n}{d} \implies H = \langle g^{\frac{n}{d}} \rangle$  which unique describes a group □

**Definition 1.5.11** (Euler's Totient Function).

For  $n \in \mathbb{N}$ , define  $\varphi(n) = \left| \{1 \leq m \leq n \mid \gcd(m,n) = 1\} \right|$

**Theorem 1.5.12.**

*Suppose  $G$  is a cyclic group of order  $n$  and  $d \mid n$  then the number of elements of  $G$  of order  $d$  is  $\varphi(d)$*

*Proof.* Assume  $G$  is cyclic group of order  $n$  and  $d \mid n$

Let  $H \leq G$  be the unique subgroup s.t.  $|H| = d \iff$  Theorem 1.5.10.II

Let  $H = \langle h \rangle \iff$  Theorem 1.5.10.I

Let  $x \in G$  have order  $d$

Consider:  $\langle x \rangle$  since  $x$  is order  $d \implies |\langle x \rangle| = d$

Since  $H$  is unique  $\implies \langle h \rangle = \langle x \rangle \implies x = h^m$  where  $\gcd(m, d) = 1 \iff$  Theorem 1.5.9

The number of choices of  $m$  is  $\varphi(d)$

□

### 1.5.1 Exercises

**Example 1.5.13.** Find all generators of  $\mathbb{Z}_{15}$  and  $\mathbb{Z}_{12}$

By Theorem 1.5.9 this implies all the numbers that are relatively prime to 12 and 15

Thus, the generators for  $\mathbb{Z}_{15}$  are 1,2,4,7,8,11,13,14

Thus, the generators for  $\mathbb{Z}_{12}$  are 1,5,7,11

**Example 1.5.14.** Show that  $\mathbb{Q}_{\geq 0}^{\times}$  is not cyclic

Assume  $\mathbb{Q}_{\geq 0}^{\times}$  is cyclic thus there exists a  $q = \frac{a}{b} \in \mathbb{Q}_{\geq 0}^{\times}$  s.t.  $\mathbb{Q}_{\geq 0}^{\times} = \langle q \rangle$

Consider:  $p$  a prime bigger than  $a, b$

Let  $\frac{1}{p} = \frac{a^m}{b^m}$

Case 1:  $m \geq 1$

$\implies b^m = pa^m \implies p \mid b^m \implies \Leftarrow$  since we picked  $p$  to be bigger than  $a, b$

Case 2:  $m \leq -1$

$\implies a^m = pb^m \implies p \mid a^m \implies \Leftarrow$

## 1.6 Homomorphism and Isomorphism

**Definition 1.6.1.** Let  $(G_1, *)$  and  $(G_2, \circ)$  be groups. A map  $\varphi : G_1 \rightarrow G_2$  is called a homomorphism if for all  $a, b, c \in G_1$  we have  $\varphi(a * b) = \varphi(a) \circ \varphi(b)$

If, in addition,  $\varphi$  is a bijection we say  $\varphi$  is an isomorphism and  $G_1$  and  $G_2$  are isomorphic

$$G_1 \cong G_2$$

**Theorem 1.6.2.** Suppose  $G = \langle g \rangle$  is a cyclic group.

If  $|G| = \infty$  then  $G \cong \mathbb{Z}$

If  $|G| = n$  then  $G \cong \mathbb{Z}_n$

*Proof.* Assume  $|G| = \infty$  Define:

$$\begin{aligned} \varphi : G &\rightarrow \mathbb{Z} \\ g^m &\mapsto m \end{aligned}$$

$$\begin{aligned} \varphi^{-1} : \mathbb{Z} &\rightarrow G \\ m &\mapsto g^m \end{aligned}$$

Let  $g^a, g^b \in G$  and  $c, d \in \mathbb{Z}$

Consider:  $\varphi(g^a g^b) = \varphi(g^{a+b}) = a + b = \varphi(g^a) + \varphi(g^b)$

Consider:  $\varphi^{-1}(c + d) = g^{c+d} = g^c g^d = \varphi^{-1}(c) + \varphi^{-1}(d)$

Consider:  $\varphi \circ \varphi^{-1}(c) = \varphi(g^c) = c$

$\varphi^{-1} \circ \varphi(g^a) = \varphi^{-1}(a) = g^a$

Thus,  $\varphi$  is a homomorphism and bijective

$$\implies G \cong \mathbb{Z}$$

Assume  $|G| = n$

$$\begin{aligned} \varphi : G &\rightarrow \mathbb{Z}_n \\ g^m &\mapsto [m] \end{aligned}$$

$$\begin{aligned} \varphi^{-1} : \mathbb{Z}_n &\rightarrow G \\ [m] &\mapsto g^m \end{aligned}$$

homomorphism: ✓

Consider:  $\varphi \circ \varphi^{-1}([c]) = \varphi(g^c) = [c]$

$\varphi^{-1} \circ \varphi(g^a) = \varphi^{-1}([a]) = g^a$

Thus,  $\varphi$  is a homomorphism and bijective

$\implies G \cong \mathbb{Z}_n$

□

**Theorem 1.6.3** (Cayley's Theorem). *Every group is isomorphic to a permutation group.*

*Proof.* Let  $G$  be a group and by Lemma 1.3.2  $\implies S_G$  is a group

Define:

$$\varphi : G \rightarrow S_G$$

$$g \mapsto l_g$$

$$l_g : G \rightarrow G$$

$$x \mapsto gx$$

- **Injective:** Let  $x, y \in G$  s.t.  $l_g(x) = l_g(y)$

$$\implies gx = gy \implies x = y$$

- **Surjective:** Let  $y \in G$

$$\text{Consider: } l_g(g^{-1}y) = gg^{-1}y = y$$

Thus  $l_g \in S_G$

Consider:  $a, b \in G$

$$\varphi(ab)(x) = l_{ab}(x) = abx = a(bx) = l_a \circ l_b(x) = \varphi(a)\varphi(b)(x)$$

**Injective:**  $a, b, x \in G$  s.t.  $\varphi(a) = \varphi(b)$

$$l_a = l_b \implies l_a(x) = l_b(x) \quad \forall x \in G \implies ax = bx \implies ae = be \implies a = b$$

$$\implies G \cong \text{im}(G) \leq S_G$$

□

**Theorem 1.6.4.** Suppose  $\varphi : G_1 \rightarrow G_2$  is a homomorphism then

I.  $\varphi(e_1) = e_2$

II.  $\forall n \in \mathbb{Z}, \varphi(g^n) = [\varphi(g)]^n$

III.  $ab = ba \implies \varphi(a)\varphi(b) = \varphi(b)\varphi(a)$

IV.  $\text{ord}(\varphi(g)) \mid \text{ord}(g)$

V.  $H \leq G_1 \implies \varphi(H) \leq G_2$

VI.  $K \leq G_2 \implies \varphi^{-1}(K) \leq G_1$

I. *Proof.*  $e_2\varphi(e_1) = e_2\varphi(e_1e_1) = e_2\varphi(e_1)\varphi(e_1) \implies e_2 = \varphi(e_1)$  □

II. *Proof.* Base Case:  $n = 0$   $\varphi(g^0) = \varphi(e_1) = e_2 = [\varphi(g)]^0$

Assume statement is true for  $k \geq 0$  for  $k \in \mathbb{N}$

Consider:  $k + 1$

$$\varphi(g^{k+1}) = \varphi(g)\varphi(g^k) = \varphi(g)[\varphi(g)]^k = [\varphi(g)]^{k+1}$$

Note:  $e_2 = \varphi(e_1) = \varphi(gg^{-1}) = \varphi(g)\varphi(g^{-1}) \implies [\varphi(g)]^{-1} = \varphi(g^{-1})$

This proves the statement for all negative integers since we just apply the induction on  $g$  then find its inverse □

III. *Proof.*  $\varphi(a)\varphi(b) = \varphi(ab) = \varphi(ba) = \varphi(b)\varphi(a)$  □

IV. *Proof.* Set  $n = \text{ord}(g)$

$$\implies \varphi(g)^n = \varphi(g^n) = \varphi(e_1) = e_2 \implies \text{ord}(\varphi(g)) \mid \text{ord}(g)$$
 □

V. *Proof.* Let  $x, y \in \varphi(H)$  s.t.  $\varphi(a) = x$  and  $\varphi(b) = y$  for  $a, b \in H$

$$\implies ab^{-1} \in H \implies \varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} \in \varphi(H)$$

Thus, by Theorem 1.4.3  $\varphi(H) \leq G_2$  □

VI. *Proof.* Let  $a, b \in \varphi^{-1}(K) \implies \varphi(a), \varphi(b) \in K$

$$\implies \varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} \in K \implies ab^{-1} \in \varphi^{-1}(K)$$

Thus, by Theorem 1.4.3  $\varphi^{-1}(K) \leq G_1$  □

**Theorem 1.6.5.** *thm:isomorphismgeneral Suppose  $\varphi : G_1 \rightarrow G_2$  is an isomorphism then  $\varphi^{-1}$  is an isomorphism*

*II.  $G_1$  is abelian  $\iff G_2$  is abelian*

*III.  $G_1$  is cyclic  $\iff G_2$  is cyclic*

*IV.  $\text{ord}(g) = \text{ord}(\varphi(g))$*

I.  $\checkmark$

II. *Proof.* Suppose  $G_1$  is abelian and  $x, y \in G_2$

$$xy = \varphi(a)\varphi(b) = \varphi(ab) = \varphi(ba) = \varphi(b)\varphi(a) = yx$$

Do same thing for other direction  $\square$

III. *Proof.* Suppose  $G_1$  is cyclic and  $\langle g \rangle = G_1$

$$\text{Take } x \in G_2 \implies \varphi(a) = x$$

$$a = g^n \implies x = \varphi(g^n) = \varphi(g)^n \implies x \in \langle \varphi(g) \rangle \implies G_2 \subseteq \langle \varphi(g) \rangle$$

$$\text{Take } x \in \langle \varphi(g) \rangle \implies x = \varphi(g)^n$$

$$\implies x = \varphi(g^n) \implies x \in G_2 \implies \langle \varphi(g) \rangle \subseteq G_2$$

$$\implies G_2 = \langle \varphi(g) \rangle \implies G_2 \text{ is cyclic}$$

Do same thing for other direction  $\square$

IV. *Proof.* By Theorem 1.6.4.IV  $\text{ord}(g) \mid \text{ord}(\varphi(g))$  and  $\text{ord}(\varphi(g)) \mid \text{ord}(g) \implies \text{ord}(g) = \text{ord}(\varphi(g))$

$\square$

### 1.6.1 Exercises

**Example 1.6.6.** Describe all homomorphisms  $\varphi : \mathbb{Z}_{12} \rightarrow \mathbb{Z}_{20}$

$$\text{ord}(1) = 12 \text{ in } \mathbb{Z}_{12} \implies \text{ord}(\varphi(1)) \mid 12 \implies \text{ord}(\varphi(1)) \in \{1, 2, 3, 4, 6, 12\}$$

$$\text{ord}(m) = \frac{n}{\gcd(m,n)} \text{ in } \mathbb{Z}_n$$

$$\bullet \text{ ord}(\varphi(1)) = 1 \implies \varphi(1) = 0 \implies \varphi(n) = 0$$

$$\bullet \text{ ord}(\varphi(1)) = 2 \implies \varphi(1) = 10$$

$$\text{this is the only element with a gcd of 10 with 20} \implies \varphi(n) = 10n$$

$$\bullet \text{ ord}(\varphi(1)) = 3, 6, 12 \text{ there is no element with these orders since these are not divisors of 10}$$

$$\bullet \text{ ord}(\varphi(1)) = 4 \implies \varphi(1) = 5, 15 \implies \varphi(n) = 5n, 15n$$

**Example 1.6.7.** Show that  $\mathbb{R} \not\cong \mathbb{R}^\times$

$$\text{Assume } \mathbb{R} \cong \mathbb{R}^\times \text{ via } \varphi : \mathbb{R}^\times \rightarrow \mathbb{R}$$

$$\text{Take } x \in \mathbb{R}^\times \text{ s.t. } \varphi(x) = -1$$

$$\text{Consider } \varphi\left(\frac{x}{2}\right)^2 = \varphi(x) = -1 \implies \varphi\left(\frac{x}{2}\right) \in \mathbb{R} \text{ with } \varphi\left(\frac{x}{2}\right)^2 = -1 \Rightarrow \Leftarrow$$

## 1.7 Cosets

**Definition 1.7.1.** Suppose  $G$  is a group  $H \leq G$ , and  $g \in G$ .

- The left coset of  $H$  with representative  $g$  is

$$gH = \{gh \mid h \in H\}$$

- The right coset of  $H$  with representative  $g$  is

$$Hg = \{hg \mid h \in H\}$$

**Example 1.7.2.**  $G = S_3, H = \{(1), (12)\}$

$$(13)H = \{(13), (123)\} = (123)H$$

$$(23)H = \{(23), (132)\} = (132)H$$

$$H(13) = \{(13), (132)\} = H(132)$$

$$H(23) = \{(23), (123)\} = H(123)$$

**Theorem 1.7.3.** Suppose that  $G$  is a group,  $H \leq G$ ,  $x, y \in G$ . The following are equivalent

I.  $xH = yH$

II.  $Hx^{-1} = Hy^{-1}$

III.  $xH \subseteq yH$

IV.  $y \in xH$

V.  $x^{-1}y \in H$

*Proof.* (I.  $\implies$  II.)

Assume I. Let  $g \in Hx^{-1} \implies g = hx^{-1}$  for  $h \in H \implies g^{-1} = xh^{-1} \in xH = yH$

$\implies g^{-1} = yj$  for  $j \in H \implies g = j^{-1}y^{-1} \in Hy^{-1} \implies Hx^{-1} \subseteq Hy^{-1}$

Do the same thing for other direction thus  $Hx^{-1} = Hy^{-1}$

(II.  $\implies$  I.)

Assume II. Let  $g \in xH \implies g = xh$  for  $h \in H \implies g^{-1} = h^{-1}x^{-1} \in Hx^{-1} = Hy^{-1}$

$\implies g^{-1} = jy^{-1}$  for  $j \in H \implies g = yj \in Hy \implies xH$

(III.  $\implies$  IV.)

Assume III. Let  $g \in xH \subseteq yH \implies g = xh = yj$  for  $h, j \in H \implies y = xhj^{-1} \implies y \in xH$

(IV.  $\implies$  V.)

Assume IV.  $y \in xH \implies y = xh$  for  $h \in H \implies x^{-1}y = h \in H$

( $V \implies I.$ )

Assume V. Let  $g \in xH \implies g = xh$  for  $h \in H \implies g = y(y^{-1}xh) \in yH$  since  $y^{-1}x \in H$

Do the same thing for the other direction this  $xH = yH$  □

**Theorem 1.7.4.** Suppose  $H \leq G$ . The cosets of  $H$  partition  $G$ . that is, they are disjoint and they union to  $G$ .

*Proof.*

Disjoint: Suppose  $x, y \in G$  s.t.  $xH \cap yH \neq \emptyset$

Take  $g \in xH \cap yH \implies g = xh = yj$   $h, j \in H \implies x^{-1}y = hj^{-1} \in H$

Thus, by Theorem 1.7.3  $xH = yH$

Union:  $\bigcup_{x \in G} xH \subseteq G \iff xH \subseteq G$

Take  $g \in G$  by Theorem 1.7.3  $\implies g \in gH \implies g \in \bigcup_{x \in G} xH \implies G \subseteq \bigcup_{x \in G} xH$

$\implies G = \bigcup_{x \in G} xH$  □

**Definition 1.7.5.** Let  $G$  be any group with  $H \leq G$ .

Define the index of  $H$  in  $G$  to be the number of left(right) cosets of  $H$  in  $G$ , denote

$$[G : H]$$

**Theorem 1.7.6.** The number of left cosets is equal to the number of right cosets

*Proof.* Let  $\mathfrak{L}_H = \{gH \mid g \in G\}$  and  $\mathfrak{R}_H = \{Hg \mid g \in G\}$

Define:

$f: \mathfrak{L}_H \rightarrow \mathfrak{R}_H$

$gH \mapsto Hg^{-1}$

Well Defined: Suppose  $xH = yH$ ,  $f(xH) = Hx^{-1} = Hy^{-1} = f(yH)$

Injective: Suppose  $f(xH) = f(yH) \implies Hx^{-1} = Hy^{-1} \implies xH = yH$

Surjective: Let  $Hg \in \mathfrak{R}_H$ ,  $f(g^{-1}H) = Hg$

Therefore,  $|\mathfrak{L}_H| = |\mathfrak{R}_H|$  □

### 1.7.1 Exercices

**Example 1.7.7.** Find all left and right cosets of  $\langle s, r \rangle \leq D_6$

$$1. \ eH = He = H$$

$$2. \ rH = \{r, r^3, r^5, sr, sr^3, sr^5\} = Hr$$

Since these union to the full set we are done

**Example 1.7.8.** Suppose that  $[G : H] = 2$  and  $a, b \notin H$ . Show that  $ab \in H$

Since  $a \notin H \implies a^{-1} \notin H$

Since  $a^{-1}, b \notin H \implies a^{-1}H \neq H$  and  $bH \neq H$

$[G : H] = 2 \implies a^{-1}H = bH \implies$  by Theorem 1.7.3.  $\forall ab \in H$

## 1.8 Lagrange's Theorem

**Theorem 1.8.1** (Lagrange's Theorem). *If  $G$  is a finite group with  $H \leq G$  then  $|G| = [G : H]|H|$ . In particular the order of a subgroup divides the order of a finite group*

*Proof.* Set  $m = [G : H]$  and take  $g_1, g_2, \dots, g_m \in G$  so that  $g_1H, g_2H, \dots, g_mH$  are all distinct.

Also,  $|g_iH| = |H| \ \forall 1 \leq i \leq m$ . This is because you take all elements of  $H$  and add  $g_i$  to them.

Since  $G = g_1H \cup g_2H \cup \dots \cup g_mH$  and these are all disjoint by Theorem 1.7.4

By the multiplication principle,  $|G| = m|H| = [G : H]|H|$  □

**Corollary 1.8.2.** *If  $|G| < \infty$  and  $g \in G$  then  $\text{ord}(g) \mid |G|$  thus  $g^{|G|} = e$*

*Proof.* Take  $H = \langle g \rangle$  and  $\text{ord}(H) = \text{ord}(g)$  thus by Theorem 1.8.1  $\text{ord}(g) = \text{ord}(H) \mid |G|$

Thus by Lemma 1.3.9  $g^{|G|} = e$  □

**Example 1.8.3.** Suppose  $|G| = 48$  and  $K \leq H \leq G$  with  $|K| = 8$ .

What are the possible values of  $|H|$ .

By Theorem 1.8.1 we know  $|H| \mid |G| = 48$  and  $|K| = 8 \mid |H|$

Thus  $|H| = 8, 16, 24, 48$

**Example 1.8.4** (The Converse of Lagrange's Theorem is false).

**Corollary 1.8.5.** *If  $p$  is prime and  $|G| = p$  then  $G \cong \mathbb{Z}_p$*

*Proof.* Take  $e \neq g \in G$  and note  $\langle g \rangle \leq G$  also  $|\langle g \rangle| \neq 1$

but by Theorem 1.8.1  $|\langle g \rangle| \mid p$  since  $p$  is prime  $\implies |\langle g \rangle| = p \implies \langle g \rangle = G$

By Theorem 1.6.2  $G \cong \mathbb{Z}_p$  □

**Corollary 1.8.6.** *For all  $a \in \mathbb{Z}, n \in \mathbb{N}$  s.t.  $\gcd(a, n) = 1$  we have  $a^{\varphi(n)} \equiv 1 \pmod{n}$*

*Proof.* Note  $[a] \in U_n$ , also  $|U_n| = \varphi(n)$

$[a] \mid |U_n| = \varphi(n) \implies [a]^{\varphi(n)} = 1 \in U_n$

$\implies [a^{\varphi(n)}] = [1] \in U_n \implies a^{\varphi(n)} \equiv 1 \pmod{n}$  □

**Theorem 1.8.7.** *Let  $p$  be an odd prime and  $G$  a group of order  $2p$  then  $G \cong \mathbb{Z}_{2p}$  or  $G \cong D_p$*

*Proof.*

Case 1: We have  $g \in G$  s.t.  $\text{ord}(g) = 2p \implies G = \langle g \rangle \cong \mathbb{Z}_{2p}$

Case 2:  $G$  does not have elements of order  $2p$ .

Assume  $G$  does not have an element of order  $p \implies$  all non identity elements have order 2.

Take  $e \neq a \neq b \neq e \in G$

Consider:  $\{e, a, b, ab\} \leq G \implies 4 \mid 2p \implies 2 \mid p \Rightarrow \Leftarrow$

Thus, there exists a  $h \in G$  s.t.  $\text{ord}(h) = p$  and form  $H = \langle h \rangle \leq G$

$[G : H] = \frac{|G|}{|H|} = \frac{2p}{p} = 2$ . Take  $g \notin H \implies G = H \sqcup gH$

Consider:  $g^2H$

Case 2a:  $g^2H = H \implies g^2 \in H$

$\implies \text{ord}(g^2) \mid p \implies \text{ord}(g^2) = 1$  thus  $g^2 = e$  or  $\text{ord}(g^2) = p$  thus  $g^{2p} = e$

$\implies \text{ord}(g) = p$  since no element has order  $2p \implies \langle g \rangle = \langle g^2 \rangle \leq H \implies g \in H \Rightarrow \Leftarrow$

$\implies \text{ord}(g) = 2$

Case 2b:  $g^2H = gH \implies g^{-1}g^2 = g \in H \Rightarrow \Leftarrow$

Thus, anything not in  $H$  has order 2

$hg \in Hg \implies hg \notin H \implies \text{ord}(hg) = 2$

$e = (hg)^2 = hghg \implies g^{-1}h^{-1} = hg \implies gh^{p-1} = hg \implies hg^k = gh^{p-k}$

Cosets are a partition so

$G = \{e, h, h^2, \dots, h^{p-1}, g, gh, \dots, gh^{p-1}\} = \langle h, g \mid e = h^p = g^2, hg = gh^{p-1} \rangle = D_p$

□

### 1.8.1 Exercises

**Example 1.8.8.** Let  $G$  be a group of order  $pqr$  where  $p, q$  and  $r$  are distinct primes. If  $H, K \leq G$  s.t.  $|H| = pq$  and  $|K| = qr$ , show that  $|H \cap K| = q$

Since  $H \cap K \leq H$  and  $H \cap K \leq K$

$$|H \cap K| \mid |H| = qp \text{ and } |H \cap K| \mid |K| = qr$$

$$\implies |H \cap K| = 1 \text{ or } q$$

Case 1:  $|H \cap K| = q$

Case 2:  $|H \cap K| = 1$

Consider:  $HK = \{hk \mid h \in H, k \in K\}$

$|HK| = |H \times K|$  however we are over counting here if  $hk$  is the same element as another  $h'k'$

$$h_1k_1 = h_2k_2 \implies h_2^{-1}h_1 = k_2k_1^{-1} \in H \cap K \text{ thus we over count by } |H \cap K|$$

$$\text{Thus } |HK| = \frac{|H||K|}{|H \cap K|} = pq^2r > pqr \Rightarrow \Leftarrow$$

$$\text{Thus } |H \cap K| = q$$

## 1.9 Normal Subgroups and Quotient Groups

**Definition 1.9.1.** Given a group  $G$ , we say  $N$  is a normal subgroup and write  $N \trianglelefteq G$  if for all  $g \in G, n \in N$  we have  $gng^{-1} \in N \iff gN = Ng \iff gNg^{-1} = N$

In this case we can form the quotient group  $G/N = \{gN \mid g \in G\}$  with  $(xN)(yN) = (xy)N$

**Theorem 1.9.2.** Given a group  $G$ :  $N \trianglelefteq G \iff G/N$  is a group

*Proof.* ( $\Leftarrow$ ) Assume  $G/N$  is a group

Let  $a, b, c, d \in G$  s.t.  $aN = cN$  and  $bN = dN$  then  $abN = cdN$

$\implies ca^{-1} = j, bd^{-1} = h, abd^{-1}c^{-1} \in N$

$ahc^{-1} = ah(ja)^{-1} = aha^{-1}j^{-1} \in N \implies aha^{-1} \in N \implies N \trianglelefteq G$

( $\implies$ ) Assume  $N \trianglelefteq G$

Let  $a, b, c, d \in G$  s.t.  $aN = cN$  and  $bN = dN \implies a \in bN \implies a = bn$  same for  $c = dh$  and  $dn = jd$  since  $dN = Nd$

$$\begin{aligned} (aN)(cN) &= (ac)N \\ &= (bndh)N \\ &= (bdjhn)N \\ &= (bd)N \\ &= (bN)(dN) \end{aligned}$$

□

### 1.9.1 Exercises

**Example 1.9.3.** Show if  $[G : H] = 2$  then  $H \trianglelefteq G$ . What is the isom. class of  $G/H$ ?

Assume  $[G : H] = 2 \implies gH \sqcup H = G$  for  $g \notin H$

Let  $x \in G$

Case 1:  $x \in H \implies xH = H = Hx$

Case 2:  $x \in gH \implies x = gh$  for  $h \in H \implies xH = ghH = gH$

Consider  $Hx$  since  $g \notin H \implies Hg = gH \implies Hx = H(gh) = g(Hh) = gH = Hx$

Therefore  $H \trianglelefteq G$  since  $|G/H| = 2$  it is isomorphic to  $\mathbb{Z}_2$